

If Else Questions

1. Largest Among Three Numbers

Write a Python program to input 3 numbers and print the largest number using if-else.

Program:

```
num1=int(input("Enter 1st number:"))
num2=int(input("Enter 2nd number:"))
num3=int(input("Enter 3rd number:"))

if num1 > num2 and num1 > num3:
    large = num1
elif num2 > num3:
    large = num2
else:
    large = num3

print("Largest number is:", large)
```

Output:

```
Enter 1st number:2
Enter 2nd number:3
Enter 3rd number:1
Largest number is: 3
```

2. Leap Year Checker

Write a program to check whether a given year is a leap year or not.

Rules:

- Divisible by 4
- But not divisible by 100
- Unless divisible by 400

Program:

```
year = int(input("Enter the year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Entered year is a leap year")
else:
    print("Entered year is not a leap year")
```

Output:

```
Enter the year: 2028
Entered year is a leap year
```

3. Electricity Bill Generator

Create a program to calculate electricity bill based on units:

Units	Rate
0-100	Rs.5/unit
101-200	Rs.7/unit
201+	Rs.10/unit

Add 18% GST at the end.

Program:

```
units = float(input("Enter the total units consumed: "))

slab1_cost = 0
slab2_cost = 0
slab3_cost = 0

if units < 0:
    print("Units consumed cannot be negative.")
elif units <= 100:
    slab1_cost = units * 5
elif units <= 200:
    slab1_cost = 100 * 5
    slab2_cost = (units - 100) * 7
else:
    slab1_cost = 100 * 5
    slab2_cost = 100 * 7
    slab3_cost = (units - 200) * 10

base_amount = slab1_cost + slab2_cost + slab3_cost
gst_amount = base_amount * 0.18
total_bill = base_amount + gst_amount

print("\n=====")
print("          ELECTRICITY BILL")
print("=====")
print(f"Total Units Consumed: {units:.2f}")
print("-----")
print(f"Slab 1 (0-100 units)   : Rs. {slab1_cost:.2f}")
print(f"Slab 2 (101-200 units): Rs. {slab2_cost:.2f}")
print(f"Slab 3 (201+ units)   : Rs. {slab3_cost:.2f}")
print("-----")
print(f"Base Amount           : Rs. {base_amount:.2f}")
print(f"GST (18%)             : Rs. {gst_amount:.2f}")
print("=====")
print(f"TOTAL PAYABLE AMOUNT  : Rs. {total_bill:.2f}")
print("=====")
```

Output:

Enter the total units consumed: 200

```
=====
                ELECTRICITY BILL
=====
Total Units Consumed: 200.00
-----
Slab 1 (0-100 units)   : Rs. 500.00
Slab 2 (101-200 units): Rs. 700.00
Slab 3 (201+ units)   : Rs. 0.00
-----
Base Amount           : Rs. 1200.00
GST (18%)             : Rs. 216.00
=====
TOTAL PAYABLE AMOUNT  : Rs. 1416.00
=====
```

4. Grade System

Input marks of 5 subjects and calculate percentage.

Print:

- A Grade -> 90+
- B Grade -> 75+
- C Grade -> 50+
- Fail -> below 50

Also print: Distinction / Pass / Fail

Program:

```
m1 = float(input("Enter marks of 1st subject : "))
m2 = float(input("Enter marks of 2nd subject : "))
m3 = float(input("Enter marks of 3rd subject : "))
m4 = float(input("Enter marks of 4th subject : "))
m5 = float(input("Enter marks of 5th subject : "))

total = m1 + m2 + m3 + m4 + m5
percentage = total * 100 / 500

print(f"percentage = {percentage:.2f}\n")

if percentage >= 90:
    print("Grade: A")
    print("Result: Distinction")
elif percentage >= 75:
    print("Grade: B")
    print("Result: First Class")
elif percentage >= 50:
    print("Grade: C")
    print("Result: Pass")
else:
    print("Grade: Fail")
```

```
print("Result: Fail")
```

Output:

```
Enter marks of 1st subject : 95
Enter marks of 2nd subject : 92
Enter marks of 3rd subject : 97
Enter marks of 4th subject : 94
Enter marks of 5th subject : 90
percentage = 93.60
```

```
Grade: A
Result: Distinction
```

5. ATM Withdrawal System

Input:

- Account balance
- Withdrawal amount

Conditions:

- Minimum balance should remain Rs.1000
- Withdrawal should be multiple of 100
- Print success/failure message

Program:

```
balance = int(input("Enter account balance: "))
withdraw = int(input("Enter withdrawal amount: "))

if withdraw % 100 != 0:
    print("Withdrawal amount should be multiple of 100")
elif balance - withdraw < 1000:
    print("Transaction failed! Minimum balance of Rs.1000 must remain")
else:
    balance = balance - withdraw
    print("Withdrawal successful!")
    print("Remaining balance = Rs." + str(balance))
```

Output:

```
Enter account balance: 1500
Enter withdrawal amount: 600
Transaction failed! Minimum balance of Rs.1000 must remain
```

6. Login Authentication System

Create a simple login system.

Correct credentials:

- Username: admin
- Password: 1234

Use nested if-else.

Program:

```
username = input("Enter Username : ")
password = input("Enter Password : ")

if username == "admin":
    if password == "1234":
        print("Login Successful")
    else:
        print("Incorrect Password")
else:
    print("Incorrect Username")
```

Output:

```
Enter Username : admin
Enter Password : 1234
Login Successful
```

7. Triangle Validator

Input 3 angles.

Check:

- Whether triangle is valid
- Then print:
 - Equilateral
 - Isosceles
 - Scalene

Program:

```
a=int(input("Enter 1st angle :"))
b=int(input("Enter 2nd angle :"))
c=int(input("Enter 3rd angle :"))

if a + b + c == 180:
    print("Triangle is valid")
    if a == b == c:
        print("Equilateral Triangle")
    elif a == b or b == c or a == c:
        print("Isosceles Triangle")
```

```
        else:
            print("Scalene Triangle")
    else:
        print("Triangle is not valid")
```

Output:

```
Enter 1st angle :90
Enter 2nd angle :50
Enter 3rd angle :40
Triangle is valid
Scalene Triangle
```

8. Employee Salary Bonus System

Input:

- Salary
- Years of experience
- Performance rating

Conditions:

- Experience > 5 years -> 20% bonus
- Rating A -> Extra 10%
- Rating B -> Extra 5%
- Otherwise no extra bonus

Print final salary.

Program:

```
salary = float(input("Enter salary: "))
experience = int(input("Enter years of experience: "))
rating = input("Enter performance rating (A/B/C): ")

bonus = 0

if experience > 5:
    bonus = bonus + (salary * 0.20)

if rating == 'A':
    bonus = bonus + (salary * 0.10)
elif rating == 'B':
    bonus = bonus + (salary * 0.05)

final_salary = salary + bonus
print(f"Final Salary = Rs.{final_salary:.2f}")
```

Output:

```
Enter salary: 100000
Enter years of experience: 6
Enter performance rating (A/B/C): A
Final Salary = Rs.130000.00
```

9. E-Commerce Discount System

Create a shopping bill program.

Conditions:

- Above Rs.5000 -> 20% discount
- Above Rs.2000 -> 10% discount
- VIP customer -> extra 5%

Add GST after discount.

Program:

```
amount = float(input("Enter shopping amount: "))
vip = input("Are you a VIP customer? (Y/N): ")

discount = 0

if amount > 5000:
    discount = discount + (amount * 0.20)
elif amount > 2000:
    discount = discount + (amount * 0.10)

if vip == 'Y' or vip == 'y':
    discount = discount + (amount * 0.05)

amount = amount - discount
gst = amount * 0.18
final_bill = amount + gst
print(f"Final Bill = Rs.{final_bill:.2f}")
```

Output:

```
Enter shopping amount: 6500
Are you a VIP customer? (Y/N): y
Final Bill = Rs.5752.50
```

10. Online Voting System

Input:

- Age
- Citizenship (Indian or not)
- Criminal record (Yes/No)

Eligible only if:

- Age >= 18
- Indian citizen
- No criminal record

Program:

```
age = int(input("Enter your age : "))
citizenship = input("Are you Indian (Y/N) : ")
criminal_record = input("Do you have any criminal record (Y/N) : ")

if age >= 18:
    if citizenship == 'Y' or citizenship == 'y':
        if criminal_record == 'N' or criminal_record == 'n':
            print("You are Eligible to vote")
        else:
            print("You are not Eligible to vote: having Criminal record")
    else:
        print("You are not Eligible to vote: Citizenship is not India")
else:
    print("You are not Eligible to vote: Age is less than 18")
```

Output:

```
Enter your age : 20
Are you Indian (Y/N) : Y
Do you have any criminal record (Y/N) : N
You are Eligible to vote
```

11. Smart Traffic Signal System

Input vehicle speed.

Conditions:

- Speed < 40 -> Safe
- 40-80 -> Normal
- 81-120 -> Warning
- Above 120 -> Challan + License Suspension

Program:

```
speed = int(input("Enter vehicle speed: "))

if speed < 40:
    print("Safe Speed")
elif speed <= 80:
    print("Normal Speed")
elif speed <= 120:
    print("Warning! Overspeeding")
else:
    print("Challan + License Suspension")
```

Output:

```
Enter vehicle speed: 70
Normal Speed
```


12. Income Tax Calculator

Calculate tax based on income slabs:

Income	Tax
Up to Rs.2.5L	0%
Rs.2.5L - Rs.5L	5%
Rs.5L - Rs.10L	20%
Above Rs.10L	30%

Add 4% cess.

Program:

```
income = float(input("Enter your income: "))

slab1 = 0
slab2 = 0
slab3 = 0

if income <= 250000:
    total = 0
elif income <= 500000:
    slab1 = (income - 250000) * 0.05
    total = slab1
elif income <= 1000000:
    slab1 = 250000 * 0.05
    slab2 = (income - 500000) * 0.20
    total = slab1 + slab2
else:
    slab1 = 250000 * 0.05
    slab2 = 500000 * 0.20
    slab3 = (income - 1000000) * 0.30
    total = slab1 + slab2 + slab3

cess = total * 0.04
total = total + cess
print(f"Total Tax = {total:.2f}")
```

Output:

```
Enter your income: 700000
Total Tax = 54600.00
```

13. Railway Ticket Booking

Input:

- Age
- Gender
- Ticket class

Conditions:

- Senior citizen discount
- Female discount
- AC class extra charge

Print final fare.

Program:

```
age = int(input("Enter age: "))
gender = input("Enter gender (M/F): ")
ticket_class = input("Enter ticket class (A for AC / N for Normal): ")

fare = 500
discount = 0

if age >= 60:
    discount = discount + (fare * 0.20)

if gender == 'F' or gender == 'f':
    discount = discount + (fare * 0.10)

fare = fare - discount

if ticket_class == 'A' or ticket_class == 'a':
    fare = fare + 200

print(f"Final Fare = Rs.{fare:.2f}")
```

Output:

```
Enter age: 70
Enter gender (M/F): f
Enter ticket class (A for AC / N for Normal): a
Final Fare = Rs.550.00
```

14. Student Result Management System

Input marks of:

- C
- Java
- DBMS
- HTML

- React

Conditions:

- Any subject below 35 -> Fail
- Otherwise calculate:
 - Percentage
 - Grade
 - Topper status (above 90%)

Use nested conditions.

Program:

```
c = float(input("Enter marks of C: "))
java = float(input("Enter marks of Java: "))
dbms = float(input("Enter marks of DBMS: "))
html = float(input("Enter marks of HTML: "))
react = float(input("Enter marks of React: "))

if c < 35 or java < 35 or dbms < 35 or html < 35 or react < 35:
    print("Result: Fail")
else:
    total = c + java + dbms + html + react
    percentage = total / 5
    print(f"Percentage = {percentage:.2f}")
    if percentage >= 90:
        print("Grade: A")
        print("Topper Status: Yes")
    elif percentage >= 75:
        print("Grade: B")
        print("Topper Status: No")
    elif percentage >= 50:
        print("Grade: C")
        print("Topper Status: No")
    else:
        print("Grade: Fail")
```

Output:

```
Enter marks of C: 95
Enter marks of Java: 94
Enter marks of DBMS: 90
Enter marks of HTML: 91
Enter marks of React: 96
Percentage = 93.20
Grade: A
Topper Status: Yes
```

15. Bank Loan Eligibility System

Input:

- Salary
- CIBIL score

- Existing loan
- Age

Conditions:

- Salary > Rs.25,000
- CIBIL > 750
- Existing loan < Rs.5L
- Age between 21-60

Print:

- Eligible
- Not Eligible
- Premium Customer Loan Offer

Program:

```
salary = float(input("Enter salary: "))
cibil = int(input("Enter CIBIL score: "))
existing_loan = float(input("Enter existing loan amount: "))
age = int(input("Enter age: "))

if salary > 25000 and cibil > 750 and existing_loan < 500000 and 21 <= age <= 60:
    print("Eligible for Loan")
    if salary > 100000 and cibil > 800:
        print("Premium Customer Loan Offer")
else:
    print("Not Eligible for Loan")
```

Output:

```
Enter salary: 1500000
Enter CIBIL score: 900
Enter existing loan amount: 250000
Enter age: 22
Eligible for Loan
Premium Customer Loan Offer
```

For Loop Questions

1. Factorial of Number

Calculate factorial using for loop.

Example: $5! = 120$

Program:

```
num = int(input("Enter the number : "))

fact = 1
for i in range(num, 0, -1):
    fact = fact * i

print(f"Factorial of {num} is {fact}")
```

Output:

```
Enter the number : 5
Factorial of 5 is 120
```

2. Reverse Table

Print multiplication table in reverse order.

Example: $5 \times 10 = 50$, $5 \times 9 = 45$...

Program:

```
num = int(input("Enter any number : "))

for i in range(10, 0, -1):
    print(f"{num} * {i} = {num * i}")
```

Output:

```
Enter any number : 5
5 * 10 = 50
5 * 9 = 45
5 * 8 = 40
5 * 7 = 35
5 * 6 = 30
5 * 5 = 25
5 * 4 = 20
5 * 3 = 15
5 * 2 = 10
5 * 1 = 5
```

3. Count Digits

Input a number and count total digits using loop.

Program:

```
num = int(input("Enter any number : "))

count = 0
temp = num
while temp != 0:
    temp = temp // 10
    count += 1

print("Total no of digits in the given no is :", count)
```

Output:

```
Enter any number : 1234
Total no of digits in the given no is : 4
```

4. Sum of Digits

Find sum of digits of a number.

Example: 1234 -> 10

Program:

```
num = int(input("Enter any number : "))

total = 0
while num != 0:
    digit = num % 10
    total = total + digit
    num = num // 10

print("Sum of digits in the given no is :", total)
```

Output:

```
Enter any number : 1234
Sum of digits in the given no is : 10
```

5. Reverse Number

Reverse a number using loop.

Example: 1234 -> 4321

Program:

```
num = int(input("Enter any number : "))

rev = 0
while num != 0:
    digit = num % 10
    rev = (rev * 10) + digit
    num = num // 10

print("Reverse no is :", rev)
```

Output:

```
Enter any number : 1234
Reverse no is : 4321
```

6. Strong Number

Check whether number is a Strong Number.

Example: 145 -> Strong Number

Program:

```
num = int(input("Enter any number : "))

temp = num
total = 0

while num != 0:
    digit = num % 10
    fact = 1
    for i in range(1, digit + 1):
        fact = fact * i
    total = total + fact
    num = num // 10

if total == temp:
    print("Strong Number")
else:
    print("Not a Strong Number")
```

Output:

```
Enter any number : 145
Strong Number
```

7. Find HCF and LCM

Find HCF and LCM of two numbers using loops.

Program:

```
num1=int (input("Enter 1st number :"))
num2=int (input("Enter 2nd number :"))

for i in range(1, min(num1, num2) + 1):
    if num1 % i == 0 and num2 % i == 0:
        hcf = i

lcm = (num1 * num2) // hcf
print("HCF =", hcf)
print("LCM =", lcm)
```

Output:

```
Enter 1st number :12
Enter 2nd number :18
HCF = 6
LCM = 36
```

8. Perfect Number

Check whether a number is perfect or not.

Example: $6 = 1 + 2 + 3$

Program:

```
num = int(input("Enter a number: "))

total = 0
for i in range(1, num):
    if num % i == 0:
        total = total + i

if total == num:
    print("Perfect Number")
else:
    print("Not a Perfect Number")
```

Output:

```
Enter a number: 6
Perfect Number
```


9. Decimal to Binary Conversion

Convert decimal number to binary using loop.

Program:

```
num = int(input("Enter decimal number: "))

binary = 0
place = 1
temp = num

while temp != 0:
    rem = temp % 2
    binary = binary + (rem * place)
    place = place * 10
    temp = temp // 2

print("Binary number =", binary)
```

Output:

```
Enter decimal number: 9
Binary number = 1001
```

10. Menu Driven Calculator

Create calculator using

- for,
- switch
- loops.

Features:

- Addition
- Subtraction
- Multiplication
- Division
- Continue option

Program:

```
while True:
    print("\n----- Calculator Menu -----")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    print("5. Exit")

    choice = int(input("Enter your choice: "))
```

```

if choice == 5:
    print("Calculator Closed")
    break

num1 = int(input("Enter 1st number: "))
num2 = int(input("Enter 2nd number: "))

match choice:
    case 1:
        print(f"Addition = {num1 + num2:.2f}")

    case 2:
        print(f"Subtraction = {num1 - num2:.2f}")

    case 3:
        print(f"Multiplication = {num1 * num2:.2f}")

    case 4:
        if num2 != 0:
            print(f"Division = {num1 / num2:.2f}")
        else:
            print("Division by zero is not possible")

    case _:
        print("Invalid Choice")

```

Output:

```

----- Calculator Menu -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter 1st number :3
Enter 2nd number :4
Addition = 7.00

----- Calculator Menu -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 5
Calculator Closed

```

11. Student Result System

Input marks of 5 students using loop and print:

- Total
- Percentage
- Grade

Program:

```
for i in range(1, 6):
    print(f"\nStudent {i}")
    sub1 = float(input("Enter marks of Subject 1: "))
    sub2 = float(input("Enter marks of Subject 2: "))
    total = sub1 + sub2
    percentage = total / 2
    print(f"Total = {total:.2f}")
    print(f"Percentage = {percentage:.2f}")
    if percentage >= 90:
        print("Grade: A")
    elif percentage >= 75:
        print("Grade: B")
    elif percentage >= 50:
        print("Grade: C")
    else:
        print("Grade: Fail")
```

Output:

```
Student 1
Enter marks of Subject 1: 99
Enter marks of Subject 2: 96
Total = 195.00
Percentage = 97.50
Grade: A
```

```
Student 2
Enter marks of Subject 1: 94
Enter marks of Subject 2: 93
Total = 187.00
Percentage = 93.50
Grade: A
```

```
Student 3
Enter marks of Subject 1: 94
Enter marks of Subject 2: 92
Total = 186.00
Percentage = 93.00
Grade: A
```

```
Student 4
Enter marks of Subject 1: 90
Enter marks of Subject 2: 89
Total = 179.00
Percentage = 89.50
Grade: B
```

```
Student 5
Enter marks of Subject 1: 88
Enter marks of Subject 2: 92
Total = 180.00
Percentage = 90.00
Grade: A
```

12. Employee Salary System

Input salary of multiple employees and calculate:

- Bonus
- Tax
- Final Salary

Using loops + conditions.

Program:

```
n = int(input("Enter number of employees: "))

for i in range(1, n + 1):
    salary = float(input(f"\nEnter salary of Employee {i}: "))

    if salary >= 50000:
        bonus = salary * 0.20
    else:
        bonus = salary * 0.10

    if salary >= 70000:
        tax = salary * 0.15
    else:
        tax = salary * 0.05

    final_salary = salary + bonus - tax
    print(f"Bonus = {bonus:.2f}")
    print(f"Tax = {tax:.2f}")
    print(f"Final Salary = {final_salary:.2f}")
```

Output:

```
Enter number of employees: 2

Enter salary of Employee 1: 900000
Bonus = 180000.00
Tax = 135000.00
Final Salary = 945000.00

Enter salary of Employee 2: 50000
Bonus = 10000.00
Tax = 2500.00
Final Salary = 57500.00
```

13. Online Exam Result System

Input marks of 5 subjects for multiple students.

Using loop:

- Calculate:
 - Total
 - Percentage
 - Grade
- Print Pass/Fail

Program:

```
students = int(input("Enter number of students: "))

for i in range(1, students + 1):
    print(f"\nEnter marks of Student {i}")
    s1 = float(input("C: "))
    s2 = float(input("Java: "))
    s3 = float(input("DBMS: "))
    s4 = float(input("HTML: "))
    s5 = float(input("React: "))

    total = s1 + s2 + s3 + s4 + s5
    percentage = total / 5

    print(f"\nTotal = {total:.2f}")
    print(f"Percentage = {percentage:.2f}")

    if percentage >= 90:
        print("Grade: A")
    elif percentage >= 75:
        print("Grade: B")
    elif percentage >= 50:
        print("Grade: C")
    else:
        print("Grade: Fail")

    if s1 < 35 or s2 < 35 or s3 < 35 or s4 < 35 or s5 < 35:
        print("Result: Fail")
    else:
        print("Result: Pass")
```

Output:

Enter number of students: 2

Enter marks of Student 1

C: 99

Java: 95

DBMS: 92

HTML: 90

React: 99

Total = 475.00

Percentage = 95.00

Grade: A

Result: Pass

Enter marks of Student 2

C: 89

Java: 86

DBMS: 97

HTML: 88

React: 92

Total = 452.00

Percentage = 90.40

Grade: A

Result: Pass

14. Mobile Recharge System

A shopkeeper recharges multiple customers.

Recharge Plans:

- Rs.199
- Rs.399
- Rs.599

Using loop:

- Input customer recharge
- Calculate total earnings of shopkeeper

Program:

```
customers = int(input("Enter number of customers: "))
total_earnings = 0

for i in range(1, customers + 1):
    print(f"\nCustomer {i}")
    print("Available Plans: 1.Rs.199  2.Rs.399  3.Rs.599")
    plan = int(input("Choose recharge plan amount: "))

    if plan == 199 or plan == 399 or plan == 599:
        total_earnings = total_earnings + plan
        print("Recharge Successful")
    else:
        print("Invalid Recharge Plan")

print(f"\nTotal Earnings of Shopkeeper = Rs.{total_earnings:.2f}")
```

Output:

```
Enter number of customers: 2

Customer 1
Available Plans: 1.Rs.199  2.Rs.399  3.Rs.599
Choose recharge plan amount: 399
Recharge Successful

Customer 2
Available Plans: 1.Rs.199  2.Rs.399  3.Rs.599
Choose recharge plan amount: 599
Recharge Successful

Total Earnings of Shopkeeper = Rs.998.00
```

15. Hospital Patient Temperature Monitor

A hospital records temperature of patients.

Using loop:

- Input temperature of 20 patients
- Count:
 - Normal patients
 - Fever patients (>100 degrees F)

Program:

```
normal = 0
fever = 0

for i in range(1, 21):
    temp = float(input(f"Enter temperature of Patient {i}: "))
    if temp > 100:
        fever += 1
    else:
        normal += 1

print(f"\nNormal Patients = {normal}")
print(f"Fever Patients = {fever}")
```

Output:

```
Enter temperature of Patient 1: 31
Enter temperature of Patient 2: 32
Enter temperature of Patient 3: 33
Enter temperature of Patient 4: 34
Enter temperature of Patient 5: 35
Enter temperature of Patient 6: 101
Enter temperature of Patient 7: 107
Enter temperature of Patient 8: 103
Enter temperature of Patient 9: 50
Enter temperature of Patient 10: 53
Enter temperature of Patient 11: 58
Enter temperature of Patient 12: 120
Enter temperature of Patient 13: 130
Enter temperature of Patient 14: 130
Enter temperature of Patient 15: 39
Enter temperature of Patient 16: 20
Enter temperature of Patient 17: 29
Enter temperature of Patient 18: 77
Enter temperature of Patient 19: 90
Enter temperature of Patient 20: 880

Normal Patients = 13
Fever Patients = 7
```